UK Road Safety SQL Analysis

Stats19 dataset, 2025 (provisional)

An exploratory SQL analysis of UK road collisions using the Department for Transport's Stats19 dataset for 2025. The project answers eight analytical questions using SQL joins, aggregations, conditional logic, window functions, and common table expressions.

Tools: MySQL · MySQL Workbench

Dataset: UK Road Safety Data (Stats19) — Collisions, Vehicles, Casualties files.

Dataset overview

Three CSVs from data.gov.uk were imported into a MySQL schema named road_safety:

Table	Description
collision	One row per reported collision (time, location, severity, conditions)
vehicle	One row per vehicle involved (type, manoeuvre, driver age)
casualty	One row per casualty (sex, age band, severity of injury)

Tables are joined on **collision_index** and **collision_ref_no**. Total reported collisions in scope: **14,914** (2025 provisional). Severity codes: 1 = Fatal, 2 = Serious, 3 = Slight, -1 = Missing.

Findings

1. Total collisions and severity breakdown (2025)

Severity	Count
Slight	11,581
Serious	3,213
Fatal	120

The vast majority of collisions (~78%) are slight. Fatal collisions are rare in relative terms but represent the most serious outcomes the dataset is used to monitor.

2. Overall fatality rate

Approximately **0.80%** of all reported collisions in 2025 were fatal. Translated: roughly **1 in every 125 collisions** results in a death.

3. Time-of-day pattern

Collisions peak at **5pm (1,285 collisions)** and stay high across the afternoon rush (3–6pm: ~1,200/hour). A secondary morning peak occurs at **8am (988)**. The quietest hours are 3am–5am (~100/hour). The shape mirrors commuting volume rather than any safety-specific factor.

4. Highest-fatality local authorities (within the top 10 by collision count)

The CTE selected the 10 local authorities with the most collisions (almost all London boroughs — E09xxxxx codes), then ranked them by fatal-collision rate:

Local authority (ONS code)	Total	Fatal	Fatal rate
E09000012	366	6	1.64%
E09000025	387	3	0.78%
E08000012	438	3	0.68%
E09000010	420	2	0.48%
E09000008	436	2	0.46%
E09000028	439	2	0.46%
E09000033	538	2	0.37%

E09000012 sits well above its peers at 1.64% — twice the next-highest. With only 6 fatal collisions, the confidence interval is wide, but the lead is large enough to warrant a closer look.

5. Wet vs dry road surfaces — serious-or-fatal rate

Road condition	Total	Serious + Fatal	Rate
Wet	2,028	515	25.39%
Dry	12,025	2,716	22.59%

Wet roads have a slightly higher serious-or-fatal rate (~2.8pp higher) — meaning per collision, wet conditions are marginally more dangerous. However, dry conditions account for ~85% of all collisions in absolute terms, simply because dry days are far more common.

6. Weekend nights vs weekday nights

A proxy analysis for drink-driving, restricted to night-time hours (10pm–4am). "Weekend night" = Friday 10pm to Sunday 4am.

Day type	Total	Serious + Fatal	Rate
Weekend night	662	179	27.04%
Weekday night	876	215	24.54%

Weekend nights show a meaningful ~2.5pp higher serious-or-fatal rate than weekday nights — consistent with the conventional intuition about late-night weekend driving. An initial cut that bucketed only Saturday and Sunday as weekend showed no difference; correctly including Friday-night-onwards in the weekend window surfaced the gap. A direct drink-related analysis would require joining the separate Contributory Factors file — left as a follow-up.

7. Vehicle types most over-represented in fatal collisions

Vehicle types with ≥50 collisions, ranked by fatal rate:

Vehicle type	Total	Fatal	Fatal rate
Goods 7.5 tonnes and over (HGV)	141	6	4.26%
Motorcycle over 500cc	504	17	3.37%
Goods vehicle — unknown weight	302	9	2.98%
Other vehicle	206	5	2.43%
Goods 3.5–7.5t	92	2	2.17%
…			
Car	16,215	122	0.75%
Pedal cycle	3,645	16	0.44%
Motorcycle 125cc and under	1,810	6	0.33%

The driver of fatality rate is mass and speed, not exposure alone: three of the top five are HGVs or goods vehicles. Large motorcycles (>500cc) — fast, motorway-capable, exposed riders — are the only non-goods entry in the top tier. Counter-intuitively, small motorbikes and pedal cycles sit near the bottom at 0.3–0.4%; they're typically urban and low-speed, so their collisions are mostly slight.

Cars dominate fatalities in absolute terms (122 deaths from 16,215 collisions) simply because of volume, but on a per-collision basis cars are far safer than HGVs or large motorbikes.

Caveat: the HGV ≥7.5t result is based on only 6 fatal collisions; the 4.26% has a wide error margin and may move by ±1–2pp on a different year of data.

8. Age and sex of fatal-collision casualties

176 casualties were recorded in fatal collisions in 2025.

Sex	Count	Share
Male	135	76.7%
Female	41	23.3%

Top age bands (male):

Age band	Fatal casualties
26 – 35	24
46 – 55	23
56 – 65	20
66 – 75	17
16 – 20	14

Casualties in fatal collisions skew heavily male (~77%), with the largest concentrations in working-age and older-male bands. The 16–20 male bracket appears with high count despite the small population size of that age group

— consistent with the well-documented "young male driver" risk pattern. Older bands (66+) also feature prominently, reflecting greater frailty: an older casualty is less likely to survive a collision of any severity.

Methodology and limitations

- **Provisional data.** 2025 figures are not yet finalised by DfT; counts and rates may shift slightly when the year is fully closed.
- **Drink-driving proxy.** There is no drink_related column on the collisions table. Drink-driving information lives in the separate Contributory Factors file, which was not loaded for this project. Q6 uses night-time hours as a behavioural proxy.
- **Wet/dry scope.** Q5 compares only road-surface codes 1 (dry) and 2 (wet/damp). Snow, ice, flood, oil, and mud (codes 3–7) were excluded to keep the comparison clean.
- **Small sample sizes** at the top of Q7 (HGV: n=6 fatal) and within Q4 (E09000012: n=6 fatal) mean those rates are indicative, not precise.
- **London-skewed top-10.** Q4's "top 10 by collision count" is dominated by London boroughs because London has the highest traffic volume — not necessarily the most dangerous roads. A per-capita or per-vehicle-mile normalisation would give a fairer comparison.
- **Joins.** Tables joined on collision_index + collision_ref_no. Casualty- and vehicle-level rows fan out from each collision, so COUNT(*) in joined queries counts casualties or vehicles, not collisions.

Files in this repo

File	Purpose
projectsql.sql	All eight queries, commented per question
results.xlsx	Query results — one sheet per question
README.md	Project documentation (this file)

What I'd do next

1. Load the Contributory Factors file and replace the night-time proxy in Q6 with a direct drink-related flag.
2. Translate ONS district codes to local-authority names using a lookup table (or the local_authority_district text column), so Q4's findings are presentation-ready without manual decoding.
3. Normalise Q4 by population or vehicle miles travelled, so the ranking reflects real risk rather than raw traffic volume.
4. Visualise the headline findings (vehicle-type fatal rates, hour-of-day curve, weekend-night comparison) as a small Power BI dashboard.